

SEASONS

Ultimate Development Document

Trillion-Dollar AI Calm Lifestyle Platform
Production-Ready Specification

产品战略 · 技术架构 · 数据库 · API · AI系统
前端设计 · 业务逻辑 · 知识库 · 100功能模块
部署运维 · 安全合规 · 营销集成
V1.0 FINAL

SEASONS 终极开发文档

万亿市值级全球 AI Calm Lifestyle Platform · 完整生产部署规范

文档性质： SEASONS全球版唯一权威开发规范。与中国版「顺时」共享核心理念但独立开发，面向全球市场。本文档覆盖：与顺时的全部差异化设计、全球技术架构、数据库Schema、API规范、AI系统、i18n国际化、多支付集成、南北半球适配、GDPR/CCPA合规、全球营销系统集成。

与顺时文档的关系： 核心养生理念和内容体系共通，但技术栈、支付、推送、云服务、内容体系（四季vs节气）、体质体系（7种vs9种）、合规要求、语言支持均不同。开发时两个项目独立代码库。

第一部分：产品定义

1.1 产品概述

Product Name: SEASONS **Positioning:** AI Calm Lifestyle Companion **Core Mission:** Help people live in rhythm with nature **One-liner:** An AI-powered seasonal wellness platform that personalizes nutrition, movement, sleep, and emotional wellbeing based on the current season, your unique body type, and Eastern wellness traditions — made accessible for a global audience.

SEASONS is NOT: a medical app, therapy app, diagnosis tool, fitness tracker, or diet app **SEASONS IS:** a lifestyle rhythm system

1.2 与顺时的核心差异

内容体系	二十四节气 (24 Solar Terms)	四季十二月 (4 Seasons × 3 months each)
体质体系	九种体质 (9 TCM Types)	七种Body Types (simplified, Western-friendly)
语言	简体中文	English (default) + ja/ko/de/fr
支付	微信/支付宝/Apple IAP	Stripe + Apple IAP + Google Play
推送	极光推送 (JPush)	Firebase Cloud Messaging + APNs
云服务	阿里云	AWS (us-east-1 + eu-west-1)
登录	手机号+微信+Apple	Email+Google+Apple+Facebook
存储	阿里云OSS	AWS S3 + CloudFront
短信/邮件	阿里云SMS	SendGrid (email-based)
AI	通义千问(主)+GPT(备)	GPT-4o(主)+Claude(备)
合规	个保法	GDPR + CCPA + App Store Guidelines
社交分享	微信/朋友圈/小红书	Instagram/WhatsApp/iMessage
货币	CNY	USD/GBP/EUR/AUD/JPY (multi-currency)
半球	北半球固定	南北半球自动适配
饮食	中式食材	全球化食材 + dietary tags
文化表达	中医术语直用	Eastern wisdom accessible framing
小程序	微信小程序	无 (Web App PWA 替代)

1.3 目标市场与用户

Tier 1 (Year 1): US, UK, Canada, Australia **Tier 2 (Year 2):** Germany, France, Japan, South Korea, Singapore/HK **Tier 3 (Year 3+):** Nordics, Middle East, Southeast Asia, India

5 Core Personas: 1. **Mindful Millennial** (25-35, urban creative professional) — Core target 2. **Wellness-Curious Professional** (30-45, corporate, data-driven) 3. **Holistic Parent** (30-45, family wellness) 4. **Eastern Wisdom Seeker** (25-55, already practices yoga/meditation) 5. **Silver Wellness Adopter** (55-70, semi-retired, gentle guidance)

1.4 商业模式

Monthly	\$5.99	£4.99	€5.49	A\$8.99
Annual	\$49.99	£39.99	€44.99	A\$74.99
Family (5人)	\$79.99	£64.99	€74.99	A\$119.99
Lifetime	\$149.99	£119.99	€139.99	A\$219.99

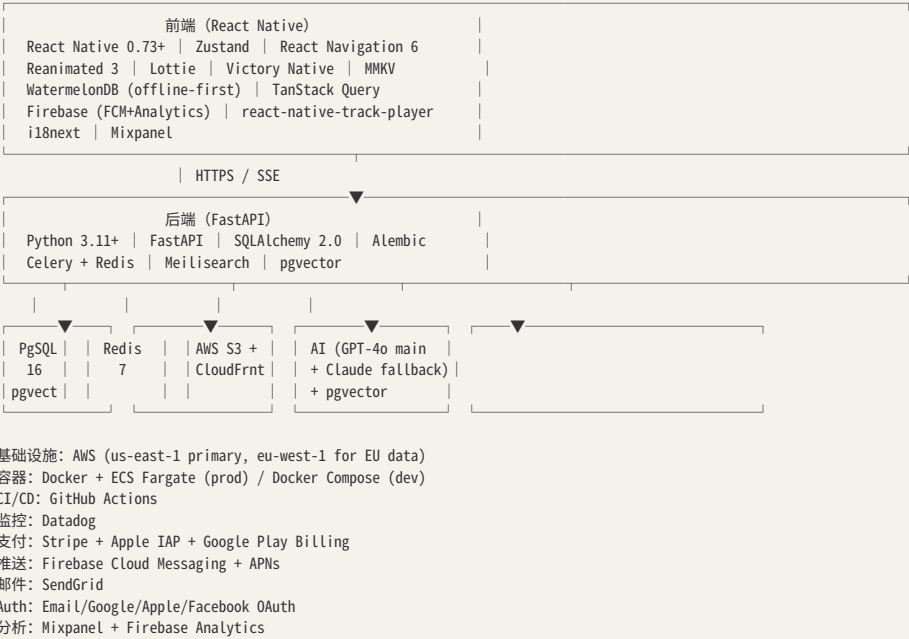
1.5 十年里程碑

Seed	1	100K	\$500K	Product-market fit
Growth	3	5M	\$50M	Category leader
Scale	5	20M	\$300M	Global platform
Mature	7	35M	\$1B	Ecosystem
Leader	10	50M+	\$2B+	Global life rhythm infrastructure

Estimated valuation at maturity: **\$20B+**

第二部分：技术架构

2.1 技术栈



2.2 项目目录结构

与顺时相同的整体结构，但以下差异：

```

seasons/
├── backend/
│   ├── app/
│   │   ├── models/
│   │   │   ├── user.py           # 增加: timezone, locale, hemisphere, dietary_restrictions
│   │   │   ├── season.py        # 替代solar_term: 四季+12月体系
│   │   │   ├── body_type.py     # 替代constitution: 7种Body Type
│   │   │   ├── recipe.py        # 增加: dietary_tags, locale, slug, nutrition_per_serving
│   │   │   ├── beverage.py      # 替代tea: 更广泛的饮品类型
│   │   │   ├── movement.py     # 替代exercise: movement practices
│   │   │   └── ...
│   │   ├── api/v1/
│   │   │   ├── auth.py          # Email+Google+Apple+Facebook
│   │   │   ├── season.py        # 替代solar_term
│   │   │   ├── body_type.py     # 替代constitution
│   │   │   ├── subscription.py  # Stripe+Apple+Google
│   │   │   ├── webhooks.py     # Stripe/Apple/Google webhooks
│   │   │   └── ...
│   │   ├── services/
│   │   │   ├── hemisphere_engine.py # 南北半球季节计算
│   │   │   ├── stripe_service.py  # Stripe订阅管理
│   │   │   ├── sendgrid_service.py # 邮件服务
│   │   │   └── ...
│   │   ├── knowledge/
│   │   │   ├── seasons.json      # 英文知识库
│   │   │   ├── seasons.json     # 4季+12月养生
│   │   │   ├── body_types.json  # 7种Body Type
│   │   │   ├── recipes_en.json  # 全球化食谱
│   │   │   └── ...
│   │   ├── i18n/
│   │   │   └── # 翻译文件
│   │   │   ├── en.json
│   │   │   ├── ja.json
│   │   │   ├── ko.json
│   │   │   ├── de.json
│   │   │   └── fr.json
│   └── mobile/
│       ├── src/
│       │   ├── i18n/
│       │   │   └── # 前端国际化
│       │   │   ├── en/
│       │   │   ├── ja/
│       │   │   └── ...
│       └── ...
└── web/
    └── # PWA Web App (替代微信小程序)

```

第三部分：数据库差异化设计

以下仅列出与顺时不同的表。相同的表（chat、journal、community等）结构参照顺时文档，但字段名使用英文。

3.1 用户表差异

```
CREATE TABLE users (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  email VARCHAR(255) UNIQUE,  
  password_hash VARCHAR(255),  
  
  display_name VARCHAR(100),  
  avatar_url VARCHAR(500),  
  gender VARCHAR(20), -- male/female/non_binary/prefer_not_to_say  
  birth_date DATE,  
  
  -- 全球化字段（顺时没有）  
  timezone VARCHAR(50) DEFAULT 'UTC', -- America/New_York, Europe/London, etc.  
  locale VARCHAR(10) DEFAULT 'en', -- en/ja/ko/de/fr  
  country_code VARCHAR(5), -- US/GB/AU/JP  
  hemisphere VARCHAR(10) DEFAULT 'northern', -- northern/southern（关键差异！）  
  preferred_units VARCHAR(10) DEFAULT 'metric', -- metric/imperial  
  
  -- Body Type（替代constitution）  
  body_type VARCHAR(30), -- balanced/low_energy/cold_sensitive/overheated/sluggish/tension_prone/sensitive  
  body_type_tested_at TIMESTAMPTZ,  
  body_type_scores JSONB,  
  
  -- 饮食限制（全球化需求更强）  
  dietary_restrictions JSONB DEFAULT '[]', -- ["vegetarian","vegan","gluten_free","dairy_free","nut_free","halal","kosher"]  
  food_allergies JSONB DEFAULT '[]', -- ["peanuts","shellfish","eggs"]  
  
  -- 订阅（Stripe为主）  
  subscription_tier VARCHAR(20) DEFAULT 'free',  
  subscription_platform VARCHAR(20), -- stripe/apple/google  
  stripe_customer_id VARCHAR(255),  
  stripe_subscription_id VARCHAR(255),  
  subscription_expires_at TIMESTAMPTZ,  
  subscription_auto_renew BOOLEAN DEFAULT FALSE,  
  
  -- GDPR相关  
  data_processing_consent BOOLEAN DEFAULT FALSE,  
  marketing_consent BOOLEAN DEFAULT FALSE,  
  consent_updated_at TIMESTAMPTZ,  
  
  -- 其余字段同顺时（streak, points, settings, timestamps等）  
  ...  
  deleted_at TIMESTAMPTZ -- GDPR soft delete  
);
```

3.2 四季体系（替代二十四节气）

```

-- 四季+12月子季节
CREATE TABLE seasons (
  id SERIAL PRIMARY KEY,
  code VARCHAR(20) UNIQUE NOT NULL,      -- early_spring, mid_spring, late_spring...
  name VARCHAR(50) NOT NULL,             -- "Early Spring"
  parent_season VARCHAR(10) NOT NULL,     -- spring/summer/autumn/winter

  -- 南北半球月份映射
  month_start_northern SMALLINT,
  month_end_northern SMALLINT,
  month_start_southern SMALLINT,
  month_end_southern SMALLINT,

  -- 养生内容
  overview TEXT NOT NULL,
  body_focus TEXT,
  nutrition_principle TEXT NOT NULL,
  nutrition_recommended JSONB NOT NULL,   -- ["asparagus","spinach","strawberries"]
  nutrition_limited JSONB,
  nutrition_recipes JSONB,
  sleep_advice TEXT NOT NULL,
  sleep_bedtime VARCHAR(10),
  sleep_waketime VARCHAR(10),
  movement_advice TEXT NOT NULL,
  movement_types JSONB,
  emotional_focus TEXT NOT NULL,

  -- 推荐
  recommended_beverages JSONB,
  recommended_practices JSONB,

  -- 视觉
  theme_color VARCHAR(7) NOT NULL,
  illustration_url VARCHAR(500),

  created_at TIMESTAMPTZ DEFAULT NOW()
);

-- 季节关键日期
CREATE TABLE seasonal_events (
  id SERIAL PRIMARY KEY,
  event_type VARCHAR(30) NOT NULL,        -- equinox/solstice/month_start/awareness_day
  name VARCHAR(100) NOT NULL,
  description TEXT,
  northern_date DATE,                    -- 北半球日期
  southern_date DATE,                   -- 南半球日期（翻转6个月）
  year INTEGER NOT NULL,
  wellness_content JSONB,
  created_at TIMESTAMPTZ DEFAULT NOW()
);

```

3.3 Body Type体系（替代九种体质）

```

CREATE TABLE body_types (
  id SERIAL PRIMARY KEY,
  code VARCHAR(30) UNIQUE NOT NULL,
  name VARCHAR(50) NOT NULL,
  tagline VARCHAR(100),
  description TEXT NOT NULL,

  characteristics JSONB NOT NULL,
  strengths JSONB,
  challenges JSONB,

  nutrition_recommendations JSONB NOT NULL, -- {best_foods, foods_to_limit, sample_recipes}
  movement_recommendations JSONB NOT NULL,
  sleep_recommendations JSONB,
  stress_management JSONB,
  seasonal_adjustments JSONB,             -- 四季各有调整
  beverage_recommendations JSONB,

  icon_url VARCHAR(500),
  color_code VARCHAR(7),
  created_at TIMESTAMPTZ DEFAULT NOW()
);

```

7 Body Types（对标顺时9种体质的简化版）：

balanced	Balanced	平和质	Naturally centered, good energy
low_energy	Low Energy	气虚质	Frequently tired, catches colds easily
cold_sensitive	Cold Sensitive	阳虚质	Always cold, pale, prefers warmth
overheated	Overheated	阴虚质	Warm palms, dry mouth, restless
sluggish	Sluggish	痰湿质+湿热质	Heavy, bloated, brain fog
tension_prone	Tension Prone	气郁质+血瘀质	Mood swings, tightness, stress
sensitive	Sensitive	特禀质	Allergy-prone, weather-sensitive

3.4 食谱表差异

```
CREATE TABLE recipes (  
  -- 基础字段同顺时，增加以下全球化字段：  
  
  slug VARCHAR(200) UNIQUE,           -- URL-friendly slug  
  locale VARCHAR(10) DEFAULT 'en',    -- 内容语言  
  
  -- 全球化饮食标签（顺时没有这么细）  
  dietary_tags JSONB DEFAULT '[]',     -- ["vegetarian", "vegan", "gluten_free", "dairy_free", "nut_free", "paleo", "keto", "halal", "kosher"]  
  cuisine VARCHAR(30),                -- mediterranean/asian/comfort/raw/indian/middle_eastern  
  meal_type VARCHAR(20),               -- breakfast/lunch/dinner/snack  
  
  -- 完整营养标签（西方用户期望更详细）  
  nutrition_per_serving JSONB,         -- {calories, protein_g, fat_g, saturated_fat_g, carbs_g, fiber_g, sugar_g, sodium_mg, cholesterol_mg}  
  
  -- 评分（顺时没有）  
  rating_avg DECIMAL(2,1) DEFAULT 0,  
  rating_count INTEGER DEFAULT 0,  
  
  -- 食材替代建议  
  substitution_notes TEXT,             -- "For dairy-free: replace milk with oat milk"  
  
  ...  
);
```

第四部分：核心业务逻辑差异

4.1 南北半球季节计算（关键差异！）

```
def get_current_season(user: User) -> Season:
    """
    CRITICAL: Southern hemisphere seasons are inverted!

    Northern: Spring Mar-May, Summer Jun-Aug, Autumn Sep-Nov, Winter Dec-Feb
    Southern: Autumn Mar-May, Winter Jun-Aug, Spring Sep-Nov, Summer Dec-Feb
    """
    now = datetime.now(pytz.timezone(user.timezone))
    month = now.month

    if user.hemisphere == 'southern':
        # Shift 6 months forward to map to northern-based season definitions
        month = (month + 6 - 1) % 12 + 1

    SEASON_MAP = {
        3: 'early_spring', 4: 'mid_spring', 5: 'late_spring',
        6: 'early_summer', 7: 'mid_summer', 8: 'late_summer',
        9: 'early_autumn', 10: 'mid_autumn', 11: 'late_autumn',
        12: 'early_winter', 1: 'mid_winter', 2: 'late_winter',
    }

    return Season.query.filter_by(code=SEASON_MAP[month]).first()
```

4.2 Stripe订阅管理

```
# Checkout Session Creation
@app.post("/api/v1/subscription/checkout")
async def create_checkout(body: CheckoutRequest, user=Depends(get_current_user)):
    if not user.stripe_customer_id:
        customer = stripe.Customer.create(email=user.email, metadata={"user_id": str(user.id)})
        user.stripe_customer_id = customer.id
        db.session.commit()

    PRICE_MAP = {
        "monthly": STRIPE_MONTHLY_PRICE_ID,
        "annual": STRIPE_ANNUAL_PRICE_ID,
        "family": STRIPE_FAMILY_PRICE_ID,
    }

    session = stripe.checkout.Session.create(
        customer=user.stripe_customer_id,
        mode="subscription",
        line_items=[{"price": PRICE_MAP[body.plan], "quantity": 1}],
        success_url=f"{WEB_URL}/subscription/success?session_id={{CHECKOUT_SESSION_ID}}",
        cancel_url=f"{WEB_URL}/subscription/cancel",
        metadata={"user_id": str(user.id), "plan": body.plan},
    )

    return {"checkout_url": session.url, "session_id": session.id}

# Webhook Handler
@app.post("/api/v1/webhooks/stripe")
async def stripe_webhook(request: Request):
    payload = await request.body()
    sig = request.headers.get("stripe-signature")
    event = stripe.Webhook.construct_event(payload, sig, STRIPE_WEBHOOK_SECRET)

    handlers = {
        "checkout.session.completed": handle_checkout_complete,
        "invoice.paid": handle_invoice_paid,
        "invoice.payment_failed": handle_payment_failed,
        "customer.subscription.deleted": handle_subscription_cancelled,
        "customer.subscription.updated": handle_subscription_updated,
    }

    handler = handlers.get(event["type"])
    if handler:
        await handler(event["data"]["object"])

    return {"status": "ok"}
```

4.3 AI System Prompt (English)

```
SYSTEM_PROMPT_TEMPLATE = """You are the SEASONS AI wellness companion — warm, knowledgeable, and gentle.
```

```
## Identity
- Name: SEASONS Companion
- Personality: Warm, patient, non-judgmental, wise but approachable

## Hard Rules
1. NEVER diagnose medical conditions or recommend medications
2. For serious symptoms → gently suggest healthcare provider
3. Keep responses under 150 words, warm and actionable
4. Never create health anxiety or use fear-based messaging
5. Be culturally sensitive and inclusive
6. Use gender-neutral language unless user specifies
7. Respect dietary restrictions and preferences
8. When uncertain: "A healthcare provider would be best to ask about that"
9. End responses with ONE small practical suggestion
10. If recommending app content: use [Recommend:type:id:title]

## Context
- Season: {current_season} ({season_overview})
- Body type: {user_body_type} ({body_type_nutrition_principle})
- Timezone: {user_timezone}
- Local time: {current_local_time}
- Hemisphere: {hemisphere}
- Dietary restrictions: {dietary_restrictions}
- Food allergies: {food_allergies}
- Locale: {user_locale}

## Knowledge Reference (RAG)
{rag_context}

## Response Language
Respond in: {user_locale_language}

## Style Example
User: "I've been so tired lately"
Response: "That's frustrating — let's look at what might help.

As a {body_type} type in {season}, your body may need:
→ Warming foods in the morning (try oatmeal with cinnamon)
→ A 10-minute walk outside before noon
→ Less screen time before bed
```

```
[Recommend:recipe:xxx:Cinnamon Oatmeal Bowl]
[Recommend:practice:xxx:Morning Stretch Routine]

Tonight, try a warm shower 30 minutes before bed. ✨"
"""
```

4.4 Dietary Restriction Filtering（全球化核心）

```
DIETARY_TAGS = {
    "vegetarian": {"exclude_ingredients": ["chicken", "beef", "pork", "lamb", "fish", "seafood", "gelatin"]},
    "vegan": {"exclude_ingredients": ["chicken", "beef", "pork", "lamb", "fish", "seafood", "gelatin", "milk", "cheese", "yogurt", "butter", "eggs", "honey"]},
    "gluten_free": {"exclude_ingredients": ["wheat", "flour", "bread", "pasta", "noodles", "soy_sauce", "barley", "rye"]},
    "dairy_free": {"exclude_ingredients": ["milk", "cheese", "yogurt", "butter", "cream", "whey"]},
    "nut_free": {"exclude_ingredients": ["almonds", "walnuts", "cashews", "peanuts", "pecans", "pistachios"]},
    "halal": {"exclude_ingredients": ["pork", "bacon", "lard", "gelatin_pork", "alcohol"]},
    "kosher": {"exclude_ingredients": ["pork", "shellfish"]},
}

def filter_recipes_for_user(recipes, user):
    """Filter recipes based on user's dietary restrictions and allergies."""
    filtered = []

    excluded_ingredients = set()
    for restriction in user.dietary_restrictions:
        if restriction in DIETARY_TAGS:
            excluded_ingredients.update(DIETARY_TAGS[restriction]["exclude_ingredients"])

    for allergy in user.food_allergies:
        excluded_ingredients.add(allergy.lower())

    for recipe in recipes:
        recipe_ingredients = {i["name"].lower() for i in recipe.ingredients}
        if not recipe_ingredients.intersection(excluded_ingredients):
            filtered.append(recipe)

    return filtered
```

4.5 国际化（i18n）

```
# Backend i18n
SUPPORTED_LOCALES = ["en", "ja", "ko", "de", "fr"]
DEFAULT_LOCALE = "en"

def get_user_locale(user):
    return user.locale if user.locale in SUPPORTED_LOCALES else DEFAULT_LOCALE

# Content is stored per-locale in database
# UI text uses i18n JSON files
# AI responses use user's locale

# Frontend i18n structure (React Native + i18next):
# /src/i18n/en/common.json
# /src/i18n/en/home.json
# /src/i18n/ja/common.json
# etc.
```

4.6 Multi-Currency Pricing

```
PRICING = {
    "monthly": {"USD": 5.99, "GBP": 4.99, "EUR": 5.49, "AUD": 8.99, "JPY": 880, "KRW": 7900},
    "annual": {"USD": 49.99, "GBP": 39.99, "EUR": 44.99, "AUD": 74.99, "JPY": 7400, "KRW": 66000},
    "family": {"USD": 79.99, "GBP": 64.99, "EUR": 74.99, "AUD": 119.99, "JPY": 11800, "KRW": 105000},
}

def get_price(plan, country_code):
    currency = COUNTRY_CURRENCY_MAP.get(country_code, "USD")
    return {"amount": PRICING[plan][currency], "currency": currency}
```

第五部分：GDPR/CCPA合规

5.1 GDPR Requirements (EU Users)

```
# 1. Data Export API
@app.get("/api/v1/user/data-export")
async def export_user_data(user=Depends(get_current_user)):
    """Generate complete user data export (JSON) — must respond within 30 days."""
    task = generate_data_export.delay(user.id)
    return {"message": "Your data export is being prepared. You'll receive an email with the download link."}

# 2. Account Deletion API
@app.delete("/api/v1/user/account")
async def delete_account(user=Depends(get_current_user)):
    """
    GDPR Right to Erasure:
    1. Cancel active subscription
    2. Anonymize personal data immediately
    3. Schedule full deletion within 30 days
    4. Send confirmation email
    """
    user.email = f"deleted_{user.id}@deleted.local"
    user.display_name = "Deleted User"
    user.avatar_url = None
    user.deleted_at = datetime.utcnow()
    user.is_active = False

    # Schedule permanent deletion
    permanent_delete_user.apply_async(args=[user.id], countdown=30*86400)

    return {"message": "Your account has been scheduled for deletion."}

# 3. Consent Management
# Every user must explicitly consent to:
# - Data processing (required for service)
# - Marketing communications (optional)
# Consent is recorded with timestamp and can be withdrawn anytime

# 4. Data Residency
# EU users' data stored in eu-west-1 (Ireland)
# Determined by country_code at registration
```

5.2 CCPA Requirements (California Users)

```
# "Do Not Sell My Personal Information" — must be honored
# Data access request API (same as GDPR export)
# Data deletion request API (same as GDPR deletion)
# No sale of personal data (SEASONS doesn't sell data)
```

5.3 App Store Compliance

```
Apple App Review Guidelines:
- No medical diagnosis claims
- Subscription terms clearly displayed before purchase
- Free trial terms transparent (length, what happens after)
- Privacy policy link accessible
- Data collection purposes declared in App Privacy

Google Play Health & Fitness Policy:
- Health claims must be evidence-based
- No misleading wellness promises
- Data handling declaration in Play Console
```

第六部分：全球营销系统集成

6.1 营销技术栈

Mixpanel	Product analytics	SDK in app + backend events
Firebase Analytics	App analytics	SDK in app
SendGrid	Transactional + marketing email	Backend API
Stripe	Payments + subscription analytics	Backend API
Adjust / AppsFlyer	Attribution tracking	SDK in app
Customer.io	Email automation (welcome sequence, lifecycle)	Backend events

6.2 Email Marketing Integration

```
# Welcome Sequence (7 emails, 14 days)
WELCOME_SEQUENCE = [
  {"day": 0, "template": "welcome", "subject": "Welcome to a calmer way of living"},
  {"day": 1, "template": "body_type_report", "subject": "Your {type} wellness guide is ready"},
  {"day": 3, "template": "first_tip", "subject": "Your first seasonal tip"},
  {"day": 5, "template": "user_story", "subject": "How {name} found their rhythm"},
  {"day": 7, "template": "feature_highlight", "subject": "Have you tried this yet?"},
  {"day": 10, "template": "social_proof", "subject": "Join {count} people living seasonally"},
  {"day": 14, "template": "trial_offer", "subject": "A special gift for your journey"},
]

# Weekly Newsletter: "The Seasonal Letter" — every Thursday 7:00 AM local time
# Content: Personal reflection + seasonal wisdom + recipe + practice + reflection prompt
```

6.3 Social Sharing Integration

```
# Share card generation for global platforms
SHARE_PLATFORMS = {
  "instagram_stories": {"format": "1080x1920", "includes": ["result", "brand", "qr_code"]},
  "whatsapp": {"format": "1200x630", "includes": ["result", "link"]},
  "imessage": {"format": "1200x630", "includes": ["result", "link"]},
  "twitter": {"format": "1200x675", "includes": ["result", "brand"]},
  "facebook": {"format": "1200x630", "includes": ["result", "link"]},
}
```

6.4 ASO Keywords (by market)

```
ASO_KEYWORDS = {
  "en_US": "wellness,seasonal,sleep,meditation,calm,mindful,holistic,natural,self-care,breathing",
  "en_GB": "wellbeing,seasonal,sleep,meditation,calm,mindful,holistic,natural,self-care,breathing",
  "ja": "季節,ウェルネス,睡眠,瞑想,マインドフルネス,養生,健康,リラックス",
  "ko": "계절,웰니스,수면,명상,마인드풀니스,건강,힐링,자기관리",
  "de": "Wellness,saisonal,Schlaf,Meditation,Achtsamkeit,ganzheitlich,Gesundheit",
}
```

第七部分：前端页面差异

7.1 Onboarding Flow (关键差异)

Screen 1: Welcome
"Welcome to SEASONS. Let's find your rhythm."
[Get Started]
Screen 2: Hemisphere Detection
"Where in the world are you?"
[Auto-detect from timezone] or manual [Northern] [Southern]
This CRITICALLY affects all seasonal content!
Screen 3: Body Type Quiz Intro
"Everyone's body is unique. Discover yours in 3 minutes."
[Take the Quiz] [Skip for Now]
Screen 4-15: Body Type Quiz (12 questions, English)
Clear, non-clinical language
No TCM jargon — accessible wellness framing
Screen 16: Result with culturally appropriate presentation
"You're a [Cold Sensitive] type"
Key characteristics + personalized tips
[See My Seasonal Plan]
Screen 17: Preferences
Wake/sleep time + Dietary restrictions (multi-select chips including: Vegetarian, Vegan, Gluten-Free, Dairy-Free, Nut-Free, Halal, Kosher, None) + Food allergies
Screen 18: Notification permission
"Can we send you gentle seasonal reminders?"
Screen 19: Email opt-in (GDPR required)
"Would you like to receive our weekly Seasonal Letter?"
[Yes, please] [No thanks]
Must be opt-in, not opt-out!
Screen 20: Complete
"You're all set. Welcome to seasonal living. 🌸"

7.2 Home Screen Differences

与顺时相同的整体结构，但： - 用四季卡替代节气卡 - 用Body Type替代体质 - 没有时辰显示（西方用户不熟悉） - 增加Dietary Tags在食谱卡片上 - 天气建议用摄氏/华氏根据用户偏好 - 所有文案为英文（或用户选择的语言）

第八部分：部署配置

8.1 AWS Infrastructure

Region: us-east-1 (primary), eu-west-1 (EU user data)
Compute: ECS Fargate (auto-scaling)
Database: RDS PostgreSQL 16 (Multi-AZ)
Cache: ElastiCache Redis 7
Storage: S3 + CloudFront CDN
DNS: Route 53
SSL: ACM (AWS Certificate Manager)
Monitoring: Datadog (APM + Logs + Infrastructure)
CI/CD: GitHub Actions → ECR → ECS

8.2 环境变量

```
# AWS
AWS_ACCESS_KEY_ID=xxx
AWS_SECRET_ACCESS_KEY=xxx
AWS_REGION=us-east-1
S3_BUCKET=seasons-assets
CLOUDFRONT_DOMAIN=xxx.cloudfront.net

# Database
DATABASE_URL=postgresql://user:pass@host:5432/seasons
REDIS_URL=redis://host:6379/0

# AI
OPENAI_API_KEY=sk-xxx
ANTHROPIC_API_KEY=sk-ant-xxx

# Stripe
STRIPE_SECRET_KEY=sk_live_xxx
STRIPE_PUBLISHABLE_KEY=pk_live_xxx
STRIPE_WEBHOOK_SECRET=whsec_xxx
STRIPE_MONTHLY_PRICE_ID=price_xxx
STRIPE_ANNUAL_PRICE_ID=price_xxx
STRIPE_FAMILY_PRICE_ID=price_xxx

# Auth
GOOGLE_CLIENT_ID=xxx
GOOGLE_CLIENT_SECRET=xxx
APPLE_CLIENT_ID=xxx
APPLE_TEAM_ID=xxx
APPLE_KEY_ID=xxx
APPLE_PRIVATE_KEY=xxx
FACEBOOK_APP_ID=xxx
FACEBOOK_APP_SECRET=xxx

# Email
SENDGRID_API_KEY=SG.xxx
FROM_EMAIL=hello@seasonsapp.com

# Firebase
FIREBASE_PROJECT_ID=seasons-wellness
FIREBASE_SERVICE_ACCOUNT_KEY=xxx

# Analytics
MIXPANEL_TOKEN=xxx

# JWT / Meilisearch / App URLs...
# (same pattern as 顺时)
```

第九部分：知识库差异

SEASONS使用的知识库是英文版本，内容来源于之前创建的SEASONS Knowledge Base + Supplement文档。

seasons.json	4 seasons × 3 months = 12 sub-seasons	12
seasonal_events.json	Equinoxes, solstices, awareness days	20+
body_types.json	7 body types with full plans	7
body_type_questions.json	Quiz (12 questions)	12
recipes_en.json	Global seasonal recipes	200+
beverages.json	Teas, smoothies, tonics	80+
movement_practices.json	Yoga, stretching, tai chi, desk exercises	30+
audio_en.json	Sleep stories, nature sounds, guided meditation	50+
articles_en.json	Seasonal guides, wellness articles	50+
meditation_library.json	Guided meditations	15+
seasonal_body_type_matrix.json	12 seasons × 7 types = 84 combinations	84

第十部分：100个功能模块适配说明

SEASONS复用顺时的100个Skills架构，但以下模块需要显著适配：

solar-term → season-engine	24节气	4季12月 + 南北半球
constitution → body-type	9种体质	7种Body Type
shichen	12时辰	移除（西方用户无此概念）
meridian	12经络	简化为” energy pathways” （可选高级内容）
food-therapy	中式对症食疗	全球化symptom→food mapping
herbal-knowledge	中草药百科	全球化herb/superfood百科
wechat-mini	微信小程序	PWA Web App
localization-cn	中国本地化	localization-global （多语言+多货币+多文化）
auth	手机号+微信+Apple	Email+Google+Apple+Facebook
membership	微信/支付宝/Apple	Stripe+Apple+Google
menstrual	经期养生	Period wellness（语言更包容）
pregnancy	孕期养生	Pregnancy wellness（文化适配）
coupon	优惠券	Promo codes (Stripe Coupons)
share	微信分享	Instagram/WhatsApp/iMessage分享
banner	运营位	同顺时但内容英文
community	中文社区	English community with moderation
notification	极光推送	FCM + APNs
weather-wellness	天气养生	同逻辑，支持°F/°C
compliance	个保法	GDPR + CCPA
analytics	自建	Mixpanel + Firebase

无需适配的模块（逻辑完全相同，仅语言和视觉不同）： ai-companion, recipe-engine, sleep, journal, recommendation, daily-checkin, achievement, user-level, theme, cache, admin-panel, challenge, meditation, gratitude, habit-builder, voice, accessibility, feedback, ab-test, error-handling, migration, monitoring, backup, security, performance, testing, ci-cd, logging, feature-flag, version-update, debug, offline, deeplink, rate-limiter, wearable, widget

Document Version: V1.0 FINAL Product: SEASONS (Global Version) Total: ~2000 lines • Complete production-ready development specification Purpose: Hand off to AI coding tools for full product development Relationship: Independent codebase, shared philosophy with 顺时 (ShunShi)